SUGGESTION - Math.

Module - 5

1. State and prove Euler's formula for a connected planar graph.

Euler's formula For a connected Planar graph:

$V - E + F = 2$ | V = no. of vertices, E = " " edges, F = " " Faces.

ProoF

Case 1: Tree → For a Tree: $E = V - 1$

also a tree has only 1 Face. $F = 1$

So, $V - E + F = V - (V-1) + 1 = V - V + 1 + 1 = 2$

hence true for Trees.

Case 2: General Connected Planar Graph

if the graph is not a tree, it contains cycle

Remove 1 edge from the cycle

- The graph remains connected.
- # edges decreased by 1.
- # faces decreased by 1.

$\therefore V - E + F = V - (E-1) + (F-1) = (V - E + F)$ Not changed.

Repeating the process eventually Reduces the graph to a tree. for which - $\boxed{V - E + F = 2}$ ✓

hence, for every connected planar graph $V - E + F = 2$ ✓

2. Prove that the number of pendent vertices in a binary tree with n vertices is $\frac{(n+1)}{2}$.

let, p = # pendent vertices.

i = # internal vertices.

in a Binary Tree, every internal vertex has exactly 2 children.

Hence total vertices: $\boxed{n = i + p}$ —— (1)

Also, no. of edges in a tree : $E = n - 1$

But each internal vertex contributes 2 edges : $\boxed{E = 2i}$ —— (2)

$$\therefore \quad 2i = n - 1 \quad (\text{From } (2))$$

$$\text{or,} \quad 2i = i + p - 1 \quad (\text{From } (1))$$

$$\text{or,} \quad \boxed{p = i + 1} \text{ —— } (3)$$

Now, $n = i + p = i + (i + 1)$ [from (3)]

$$\text{or,} \quad n = 2i + 1$$

$$\therefore \quad \boxed{i = \frac{n-1}{2}} \text{ —— } (4)$$

Now, $p = i + 1$

$$= \frac{n-1}{2} + 1 \quad [\text{From } (4)]$$

$$\therefore \quad \boxed{p = \frac{n+1}{2}} \quad (\text{Proved})$$

3. Prove that a tree with n number of vertices has $(n - 1)$ number of edges.

Base case.

For $n = 1$: a tree with 1 vertex has 0 edges : $1 - 1 = 0$

(True)

Induction Hypothesis

Assume a tree with k-vertices has $(k-1)$ edges.

Induction Step

Consider a tree with $(k+1)$ vertices

A tree always has at least 1 pendent vertex.

Remove one pendent vertex and its edge.

The remaining graph is still a tree with k-vertices.

By induction hypothesis it has $(k-1)$ edges.

Adding back the Removed edge: $(k-1) + 1 = k$

Since $(k+1)$ vertices have k-edges : $k = (k+1) - 1$

Hence proved.

4. State and prove generalized Euler theorem for planar graph.

Statement: If a planar graph has V - vertices, E - edges, F - Faces, K - connected components then -

$$\boxed{V - E + F = 1 + K}$$

Proof:

For each connected component, $V_i - E_i + F_i = 2$

Suppose there are K connected components.

Adding all equations: $\boxed{V - E + F' = 2K}$

But the outer face is common to all components.

Hence, actual no. of faces. $F = F' - (K-1)$

Substitute, $V - E + [F + (K-1)] = 2K$

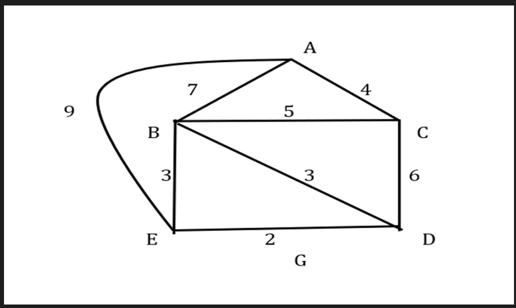
$$\boxed{V - E + F = 1 + K}$$ (Proved)

5. Prove that a connected graph with n vertices and $(n-1)$ edges is a tree.

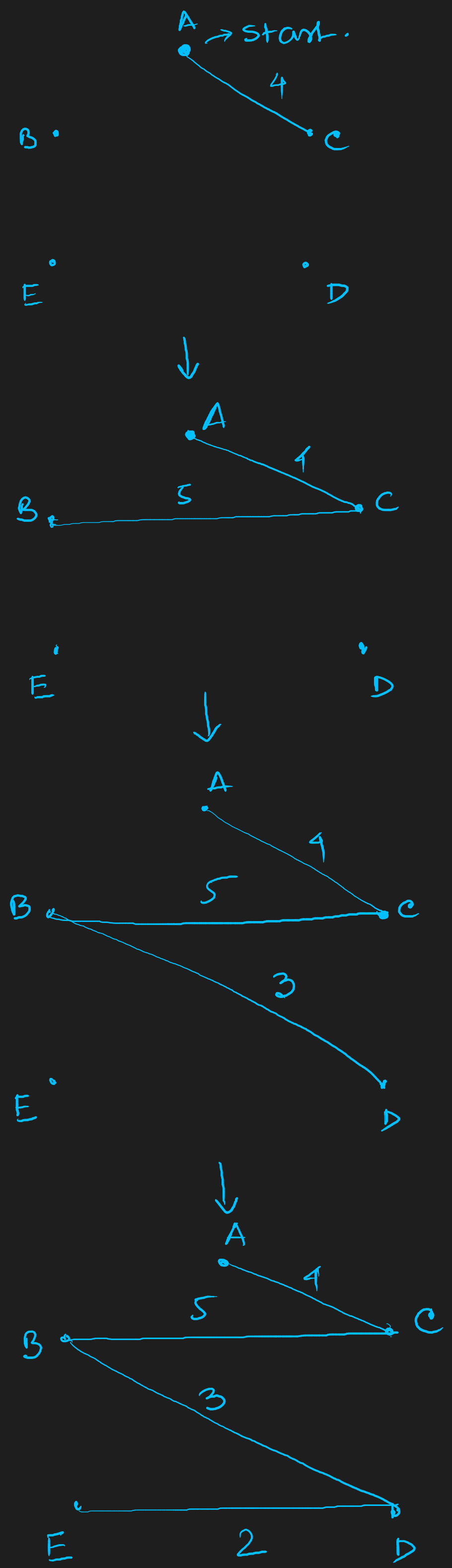
Given:

* Graph is connected
* # edges $= n-1$

7. Find a minimal spanning tree by the Prims and Kruskal algorithm of the graph G given below:

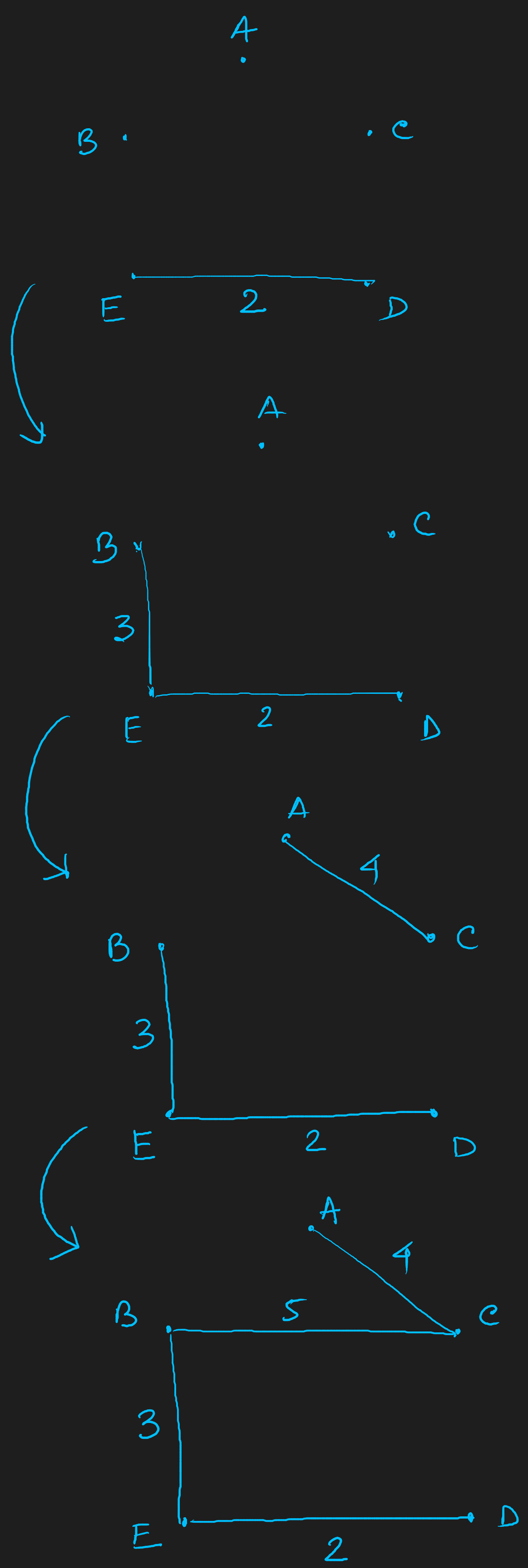


Prims.

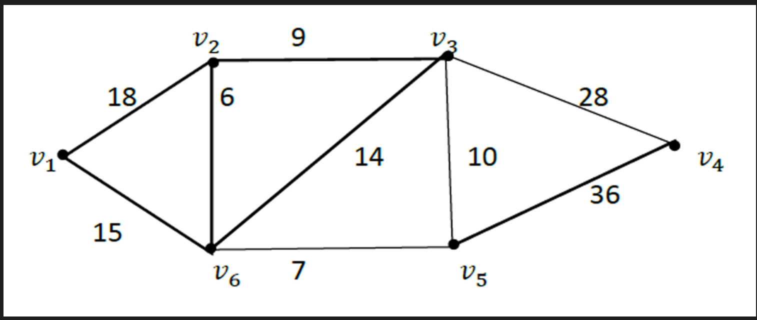


Cost = 14 ✓

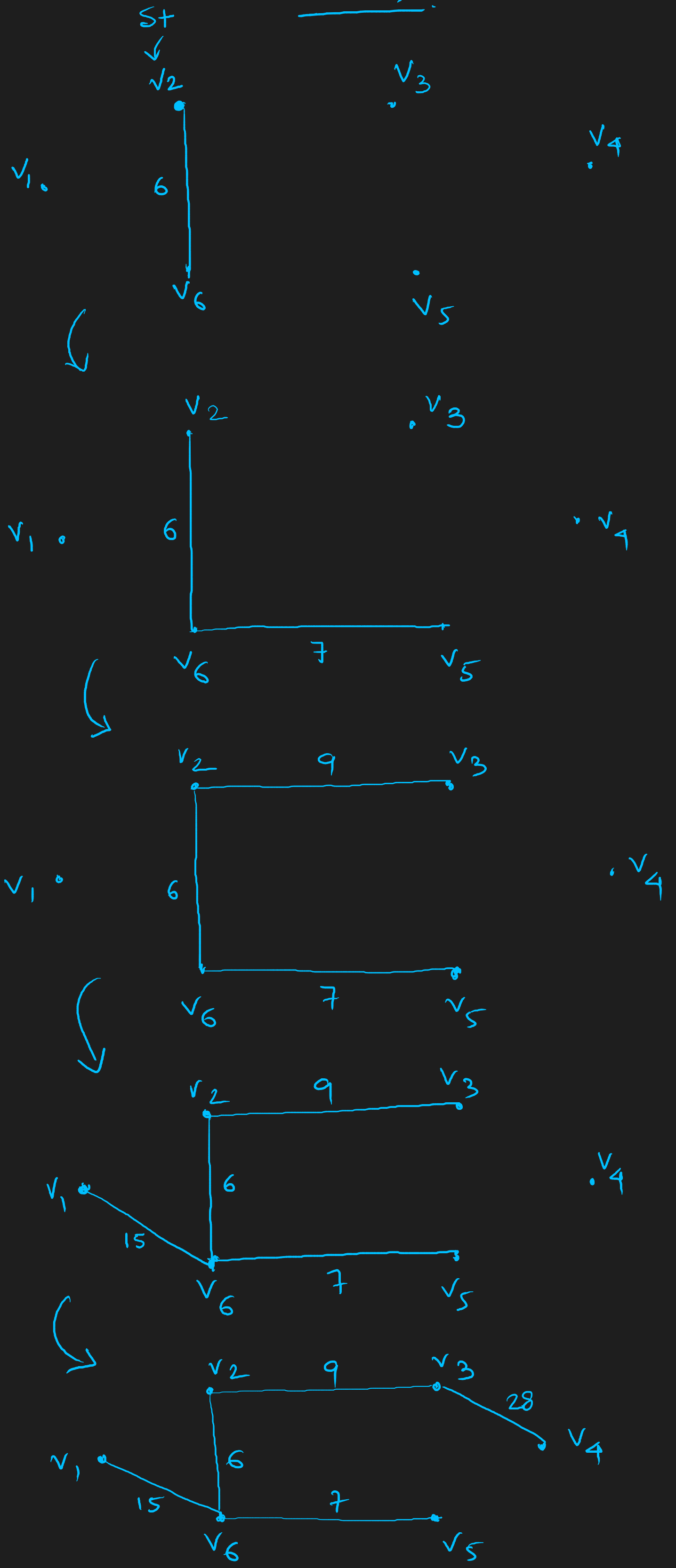
Kruskal



Cost = 4+5+3+2 = 14

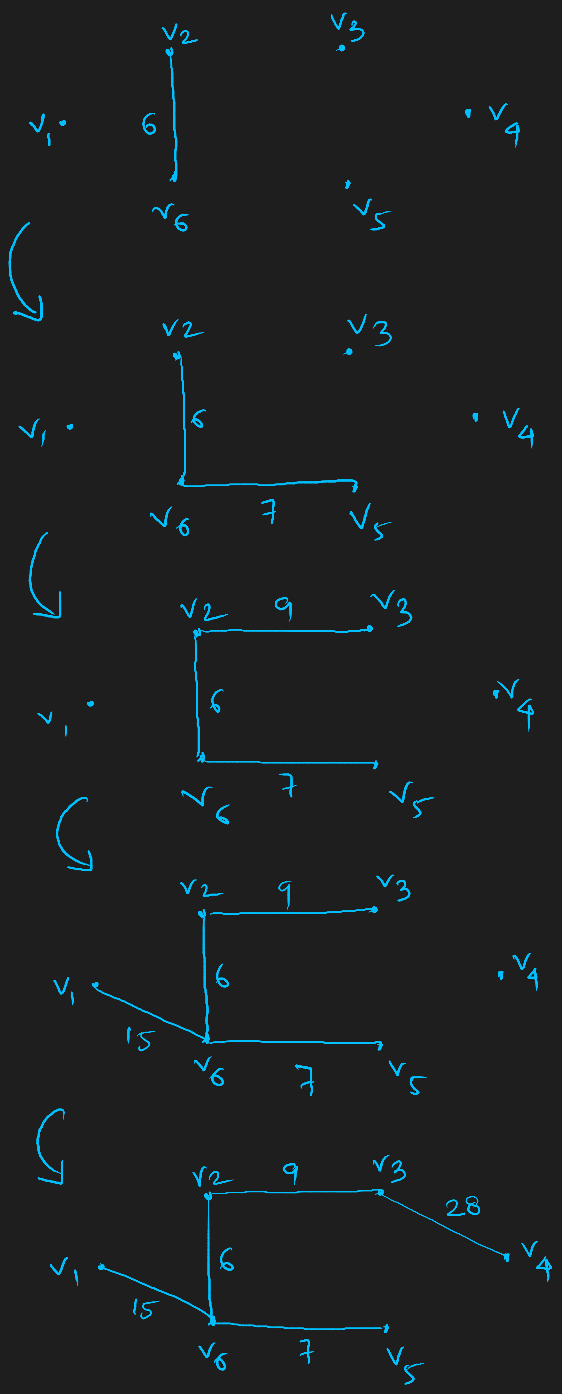


Prims.

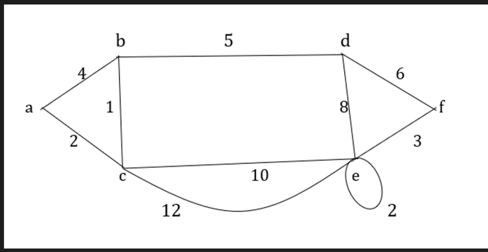


Cost = $15+9+6+7+28$
$= 65$

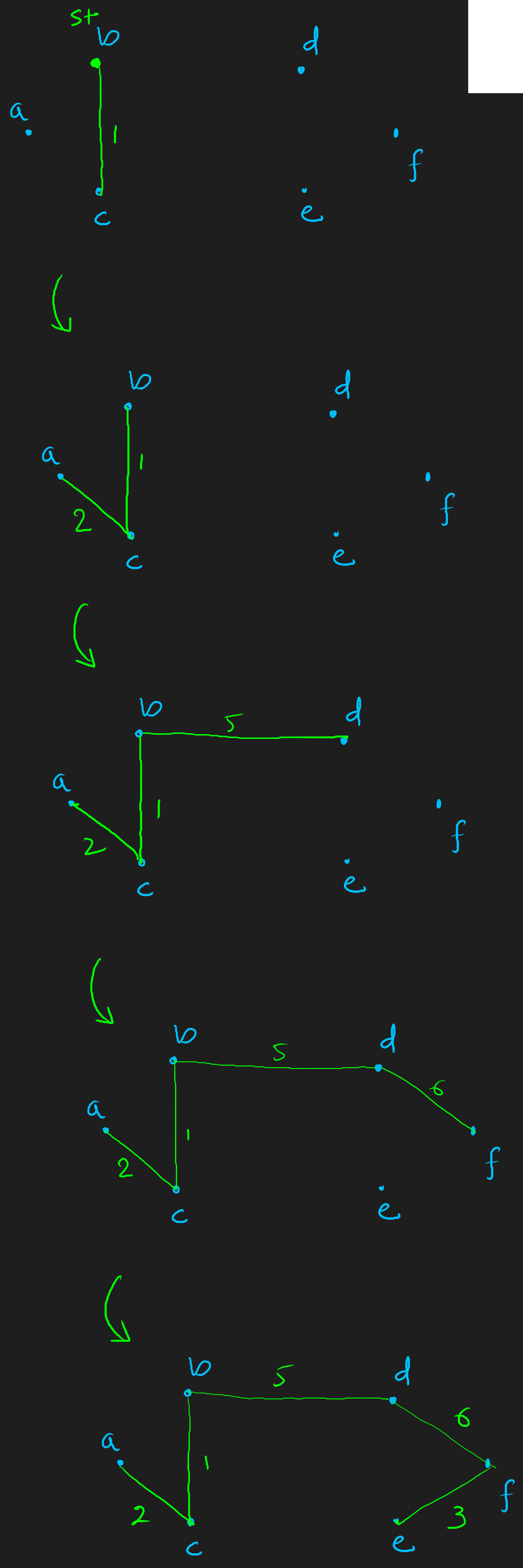
Kruskal.



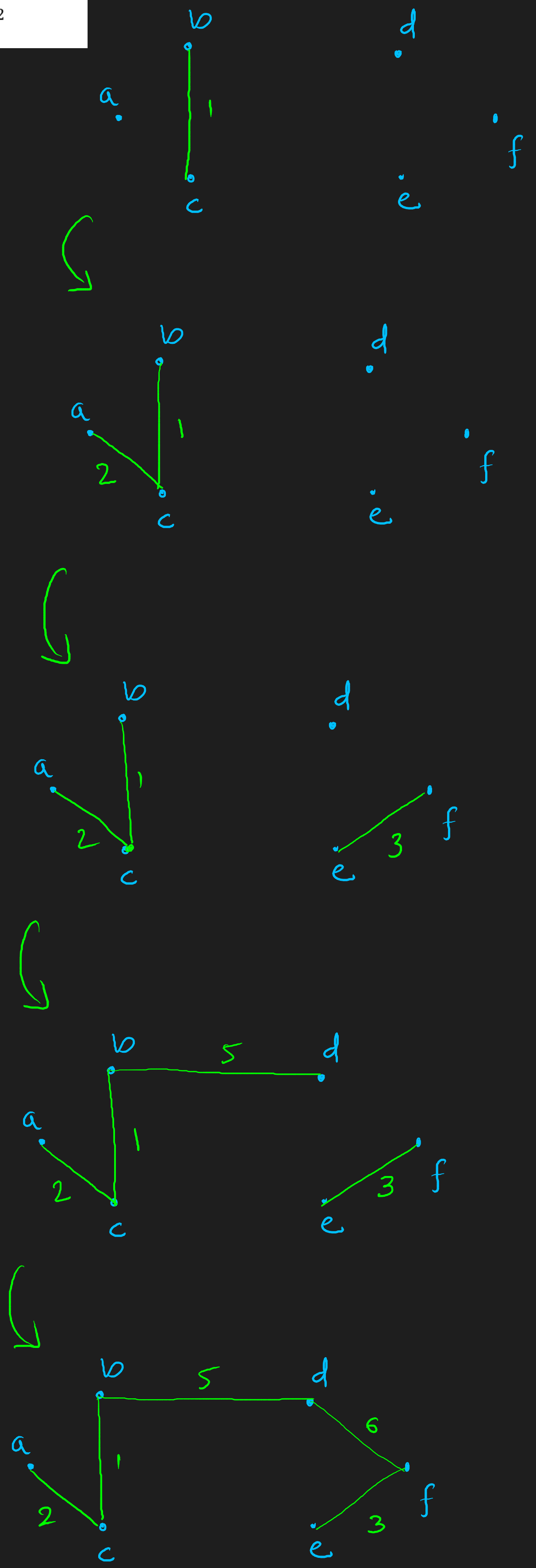
Cost = $15+6+7+9+28$
$= 65$



Prims



Kruskal



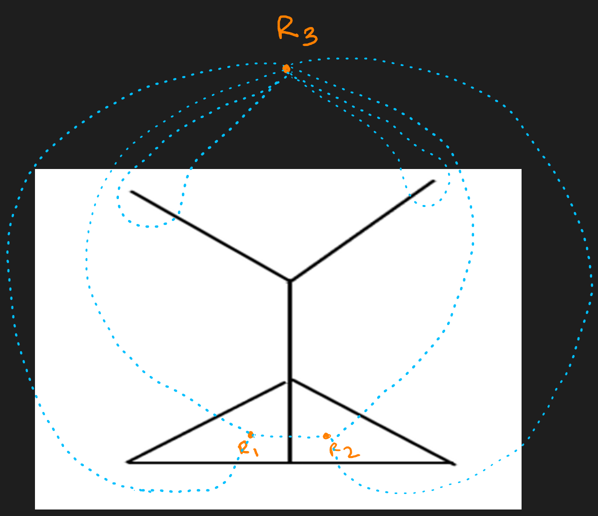
Cost = $1 + 2 + 3 + 6 + 5$

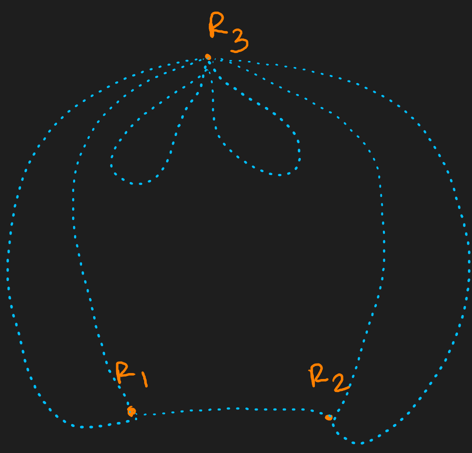
$= 17$ ✓

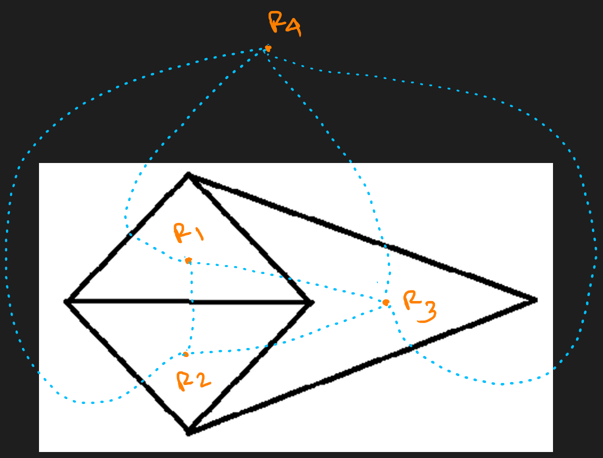
Cost = $1+2+3+5+6$

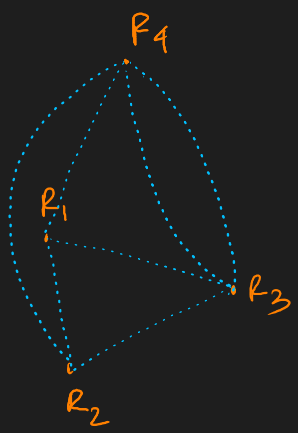
$= 17$ ✓

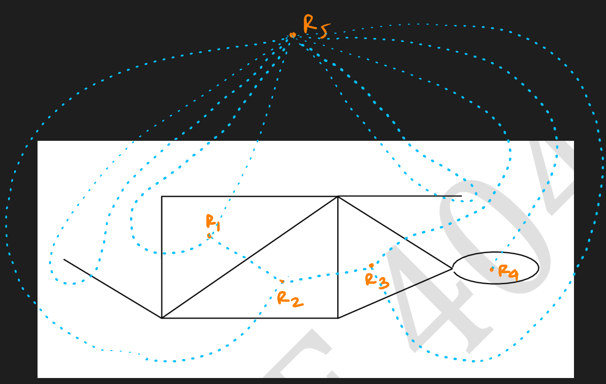
8. Draw the dual of the following graph

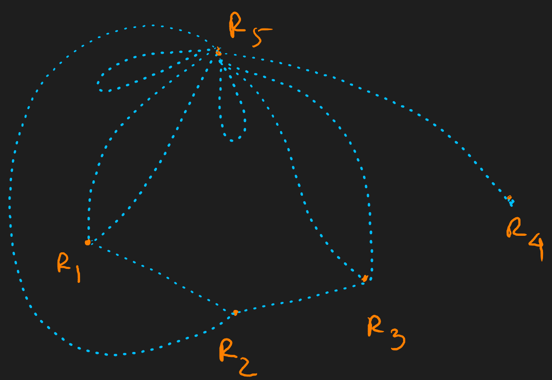


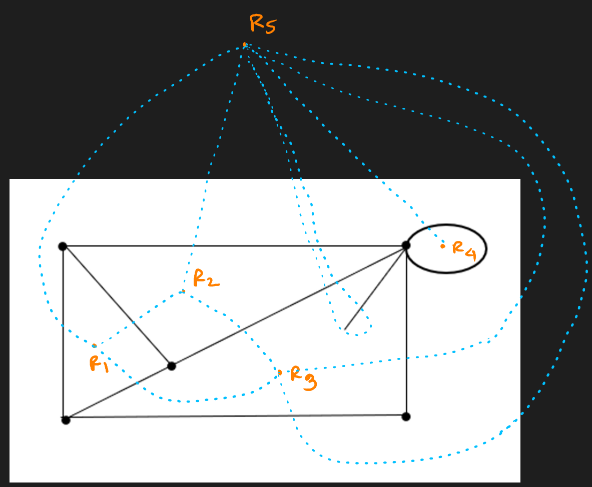


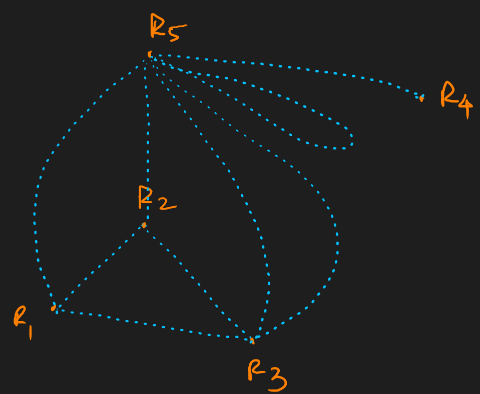




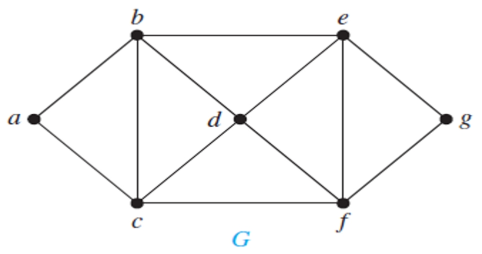








9. Find three different independent set of vertices with different size and also find the chromatic number of the graph given below



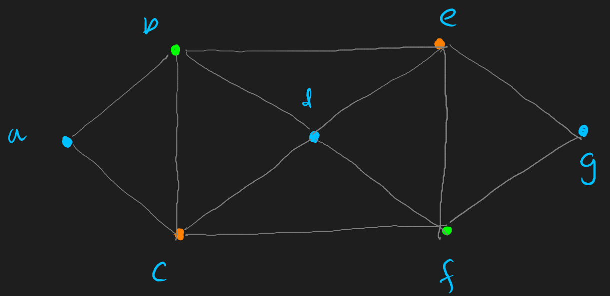
$V = [a,b,c,d,e,f,g]$

Connections:

$a \to b, c$
$b \to a, c, d, e$
$c \to a, b, d, f$
$d \to b, c, e, f$
$e \to b, d, f, g$.
$f \to c, d, g, e$
$g \to e, f$

independent sets.

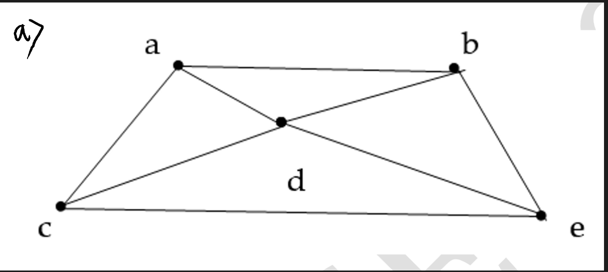
$\{a,d,g\}$, $\{b,f\}$, $\{e,c\}$ Ans.



$\to \chi(G) = 3$ Ans.

10. Find the chromatic polynomial of the given graph, hence find its chromatic number.

a)



$V = \{a, b, c, d, e\}$

Let, total colors $= k$

Start colorin form 'a'.

ways to color $a = k$
ways to color $b = k-1$
way to color $c \to \boxed{b=c} = 1$
$\to \boxed{b \neq c} = (k-2)$

ways to color $d \to \boxed{b=c} = (k-2)$
$\to \boxed{b \neq c} = (k-3)$

ways to color $e \to \boxed{b=c} = (k-2)$
$\to \boxed{b \neq c} = (k-3)$

$\therefore$ Polynomial. $P(G;K) = K(K-1)\left[1\cdot(K-2)(K-2) + (K-2)(K-3)^2\right]$

$= K(K-1)\left[(K-2)^2 + (K-2)(K-3)^2\right]$

$= (K^2-K)\left\{(K^2-4K+4) + (K-2)(K^2+9-6K)\right\}$

$= (K^2-K)\left\{(K^2-4K+4) + K^3+9K-6K^2-2K^2-18+12K\right\}$

$= (K^2-K)(K^3-7K^2+17K-14)$

$= K^5-7K^4+17K^3-14K^2-K^4+7K^3-17K^2+14K$

$$\boxed{P(G,K) = K^5-8K^4+24K^3-31K^2+14K.}$$

$P(G,1) = 1-8+24-31+14 = 0$

$P(G,2) = 32-8\times16+24\times8-31\times4+28 = 0$

$P(G,3) = 243-8\times81+24\times27-31\times9+42 = 6 > 0$ $\therefore$ $\boxed{\chi(G)=3}$

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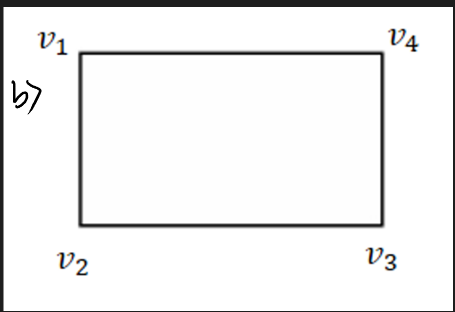
b)



$V = \{v_1, v_2, v_3, v_4\}$

Suppose total colors $= K$

start coloring from 'v_1'

ways to color $v_1 = K$

ways to color $v_2 = (K-1)$ [b/c v_1 & v_2 adjacent]

ways to color v_3 $\longrightarrow$ if take $\boxed{v_1 = v_3}$ $= 1$

$\longrightarrow$ if take $\boxed{v_1 \neq v_3}$ $= (K-2)$

ways to color v_4 $\longrightarrow$ $\boxed{v_1 = v_3}$ $\longrightarrow$ $\boxed{v_2 = v_4}$ $= 1$

$\longrightarrow$ $\boxed{v_2 \neq v_4}$ $= (K-2)$

$\longrightarrow$ $\boxed{v_1 \neq v_3}$ $\longrightarrow$ $\boxed{v_2 = v_4}$ $= 1$

$\longrightarrow$ $\boxed{v_2 \neq v_4}$ $= (K-3)$

$$\therefore \quad P(G,k) = k(k-1)\left[1.1 + 1.(k-2) + (k-2).1 + (k-2)(k-3)\right]$$

$$= (k^2-k)\left(1 + k - 2 + k - 2 + k^2 + 6 - 5k\right)$$

$$= (k^2-k)(k^2-3k+3) = k^4 - 3k^3 + 3k^2 - k^3 + 3k^2 - 3k$$

$$\boxed{P(G,k) = k^4 - 4k^3 + 6k^2 - 3k}$$

Now,

$$P(G,1) = 1 - 4 + 6 - 3 = 0$$

$$P(G,2) = 16 - 32 + 24 - 6 = 2 > 0 \qquad \therefore \boxed{\chi(G) = 2}$$

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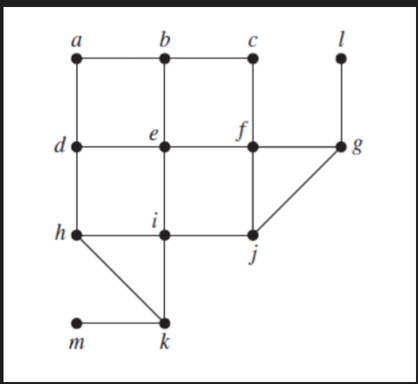
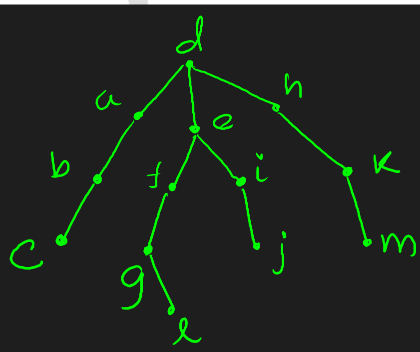
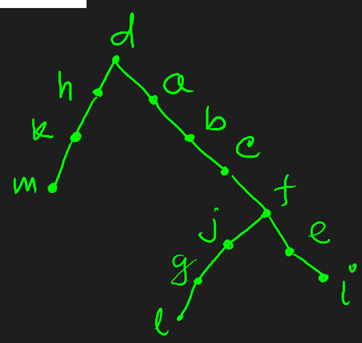
11. Find a spanning tree starting from **d** for the following graph using **breadth-first search** and **depth-first search**

(i)





BFS



DFS

Lvl 0 : d

Lvl 1 : a, e, h

Lvl 2 : b, f, i, k

Lvl 3 : c, g, j, m

Lvl 4 : l

$d \to h \to k \to m$

back track

$d \to a \to b \to c \to f$

$f \to j \to g \to l$

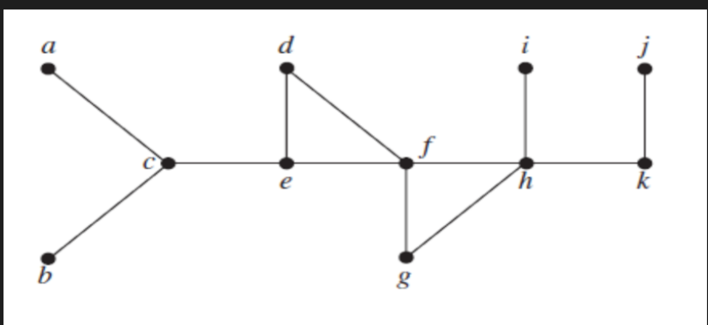
back track

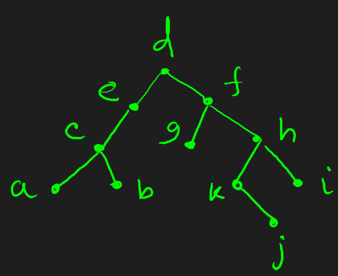
$f \to e \to i$

BFS path : d, a, e, h, b, f, i, k, c, g, j, m, l.

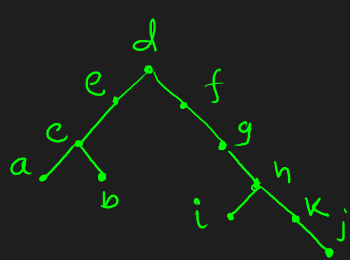
DFS path : d, h, k, m, a, b, c, f, j, g, l, e, i

(ii)





BFS



DFS

BFS

$\boxed{lvl\,0}$: d

$\boxed{lvl\,1}$: e, f

$\boxed{lvl\,2}$: c, g, h

$\boxed{lvl\,3}$: a, b, k, i

$\boxed{lvl\,4}$: j

BFS path : d, e, f, c, g, h, a, b, k, i, j

DFS

d → e → c → a

backtrack

c → b

backtrack

d → f → g → h → i

backtrack

h → k → j

DFS path : d, e, c, a, b, f, g, h, i, k, j

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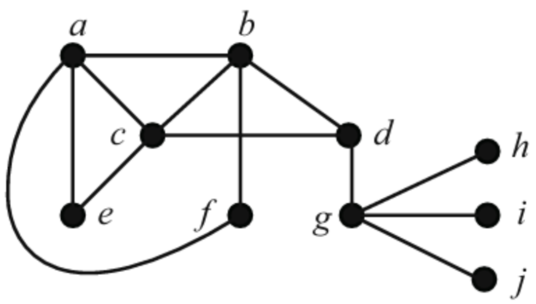
12. Find two different matchings of different size of the following graph. Also find maximum matching and maximal matching of the following graph.



Matching

Cond^n s
- degree of every vertex should be ≤ 1
- No adjacent edges.

Maximal Matching

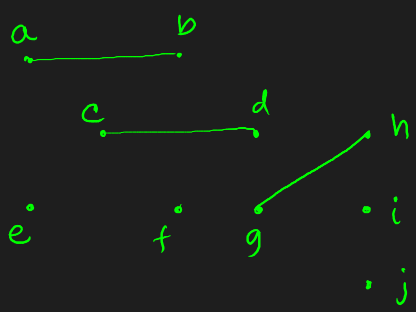
- cannot increase more edges it will make adjacent edges.

Maximum Matching.

- Max no. of edges.

Perfect Matching

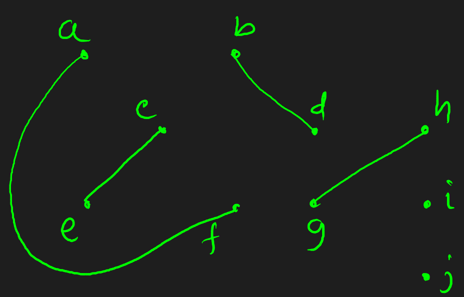
- every vertex degree = 1



Matching 1

{(a,b), (c,d), (g,h)}

Size = 3

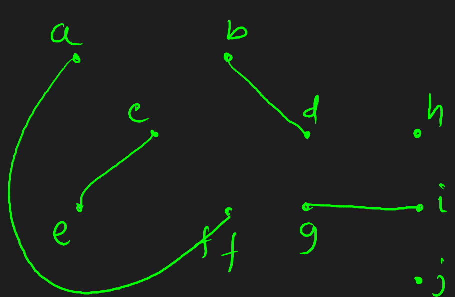


Matching 2

{(a,f), (e,c), (b,d), (g,h)}

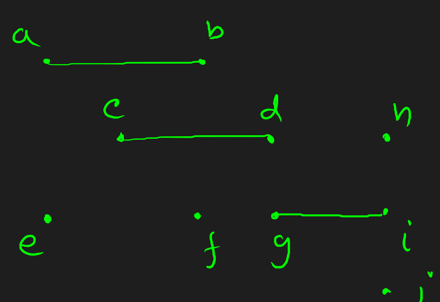
Size = 4

Maximum Matching.



{(a,f), (e,c), (b,d), (g,i)} $\nu(G) = 4$.

Maximal Matching



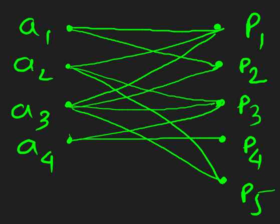
{(a,b), (c,d), (g,i)}

13. Applicants a_1, a_2, a_3, a_4 apply for five posts p_1, p_2, p_3, p_4 and p_5. The applications are done as follows: $a_1 \to \{p_1, p_2\}$, $a_2 \to \{p_1, p_3, p_5\}$, $a_3 \to \{p_1, p_2, p_3, p_5\}$ and $a_4 \to \{p_3, p_4\}$. (i) Draw a bipartite graph of the relationship between applicants and posts (ii) Is there any perfect matching of the set of applicants into the set of post? If yes use Hall's marriage theorem to show that every applicant can be offered a single post.

Applicants = $\{a_1, a_2, a_3, a_4\}$

posts = $\{P_1, P_2, P_3, P_4, P_5\}$

(i)



→ Bipartite graph

(ii) Hall's Marriage Theorem

We need to check whether $\boxed{|N(S)| \geq |S|}$ ∀ subset S of applicants.

Single Applicants.

$N(a_1) = \{P_1, P_2\}$ $\quad |N(a_1)| = 2 \geq 1$

also, $|N(a_2)| = 3$, $|N(a_3)| = 4$, $|N(a_4)| = 2$ → All satisfy Hall's condit^ns

Pairs.

$N(\{a_1, a_2\}) = \{P_1, P_2, P_3, P_5\}$ $\therefore |N(a_1 a_2)| = 4 \geq 2$

$|N(a_1 a_4)| = 4 \geq 2$, $|N(a_3 a_4)| = 5 \geq 2$ and others also ≥ 2.

→ All satisfy

Triplets

$|N(a_1 a_2 a_3)| = 4 \geq 3$

$|N(a_1 a_2 a_4)| = 5 \geq 3$ ... others also. → Satisfy all.

All

$|N(a_1 a_2 a_3 a_4)| = 5 \geq 4$ → Halls condition Satisfy for all subsets.

By Hall's Marriage theorem, a matching exists that saturates all applicants. $S \subset \{a_1, a_2, a_3, a_4\}$

Hence every applicant can be assigned exactly one post.

One such perfect matching is : $(a_1 P_2), (a_2 P_5), (a_3 P_1), (a_4 P_4)$

14. Suppose there are 3 boys a, b, c knows four girls x, y, z, w as follows:

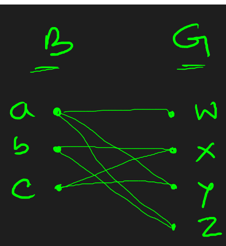
Boy	Girls
A	w, y, z
B	x, z
c	x, y

Draw a bipartite graph corresponding to the table, (ii) Find five different matchings, (iii) Show that the Hall's marriage condition is satisfied for the above matching.

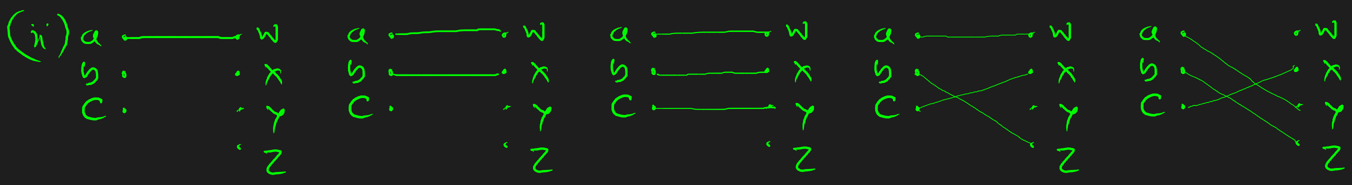
$B = \{a, b, c\}$

$G = \{w, x, y, z\}$

(i)



(ii)



$\boxed{M_1}$ (a, w)

$\boxed{M_2}$ $\{(a, w), (b, x)\}$

$\boxed{M_3}$ $\{(a, w), (b, x), (c, y)\}$

$\boxed{M_4}$ $\{(a, w), (c, x), (b, z)\}$

$\boxed{M_5}$ $\{(a, y), (c, x), (b, z)\}$

(iii) Hall Marriage Theorem $|N(S)| \geq |S|$

Single Boys	Pairs	All Boys
$\lvert N(a)\rvert = 3 \geq 1$	$N(a,b) = 4 \geq 2$	$N(a,b,c) = 4 \geq 3$
$\lvert N(b)\rvert = 2 \geq 1$	$N(a,c) = 4 \geq 2$	
$\lvert N(c)\rvert = 2 \geq 1$	$N(b,c) = 3 \geq 2$	

Since, Hall's condition holds for every subset. $\forall\, S \subseteq \{a, b, c\}$ therefore, $\exists$ a matching that matches every boy with distinct girl.

One such complete matching is

$$\{(a, w), (b, z), (c, x)\}$$